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United States Patent	12418010
Kind Code	B2
Date of Patent	September 16, 2025
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Semiconductor package with improved heat distribution

Abstract

A semiconductor structure and processes of forming the same are provided. A semiconductor structure according to the present disclosure includes a first die having a front surface and a back surface and a second die bonded to the back surface of the first die. The first die includes a plurality of trenches adjacent the back surface and the plurality of trenches are filled with a liquid.

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Appl. No.: 17/830861

Filed: June 02, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230395586 A1	Dec. 07, 2023

Publication Classification

Int. Cl.: H01L25/00 (20060101); H01L23/00 (20060101); H01L23/48 (20060101); H01L25/065 (20230101)

U.S. Cl.:

CPC H01L25/50 (20130101); H01L23/481 (20130101); H01L24/08 (20130101); H01L24/80 (20130101); H01L25/0657 (20130101); H01L2224/08145 (20130101); H01L2224/80006

(20130101); H01L2224/80895 (20130101); H01L2224/80896 (20130101);
H01L2225/06541 (20130101); H01L2225/06582 (20130101); H01L2225/06589
(20130101)

Field of Classification Search

CPC: H10B (80/00); H01L (25/0657); H01L (25/50); H01L (25/0652); H01L (24/80); H01L (24/473); H01L (2224/80006); H01L (2225/06524); H01L (2225/06527); H01L (2225/06586)

USPC: 257/777

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5313094	12/1993	Beyer	N/A	N/A
9293437	12/2015	Yu et al.	N/A	N/A
9496196	12/2015	Yu et al.	N/A	N/A
11233035	12/2021	Chen	N/A	N/A
2006/0145356	12/2005	Liu	N/A	N/A
2010/0187683	12/2009	Bakir	N/A	N/A
2016/0056089	12/2015	Taniguchi	438/456	H01L 21/30625
2019/0333893	12/2018	Yu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105210185	12/2014	CN	N/A
201032352	12/2009	TW	N/A
202010047	12/2019	TW	N/A

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Background/Summary

BACKGROUND

(1) The integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs, where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

(2) Three-dimensional (3D) packaging or chip stacking generally refers to stack and attach chips vertically to increase efficiency and make better use of available space. It can enable heterogeneous integration of several dies with different chip sizes and/or functionalities. An example of

implementation of 3D packaging is a system on integrated chip (SoIC) device or a high performance computing (HPC) device. The dense packing and vertical integration create challenges in heat dissipation as accumulation of heat may reduce the performance. While existing 3D package structures are generally adequate for their intended purposes, they are not satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a flow chart illustrating an embodiment of a method of forming a package structure, according to various aspects of the present disclosure.

(3) FIGS. 2-17 are fragmentary cross-sectional views of a workpiece undergoing operations of the method in FIG. 1, according to various aspects of the present disclosure.

DETAILED DESCRIPTION

(4) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(5) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(6) Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(7) Three-dimensional (3D) packaging or chip stacking generally refers to stack and attach chips vertically to increase efficiency and make better use of available space. It enables heterogeneous integration of several dies with different chip sizes and/or functionalities. An example of implementation of 3D packaging is a system on integrated chip (SoIC) device or a high performance computing (HPC) device. Due to vertical integration, heat generated in a 3D package is dissipated mainly through conduction of the semiconductor substrates. Manufacture of 3D package structures includes steps to thin substrates of dies to reduce the resistance of through-substrate-vias (TSVs). Without substrate thinning, it can be difficult to satisfactorily fill the high-aspect-ratio TSV openings, resulting in voids that increase resistance. However, substrate thinning also reduces heat dissipation capabilities. Without sufficient heat dissipation, the performance of the 3D package may suffer. Additionally, high power density regions in the 3D package may form hot spots, which can locally reduce carrier mobility.

(8) The present disclosure provides a package structure that increases heat dissipation through semiconductor substrates of dies in the package structure. According to the present disclosure, a plurality of trenches are formed in substrates of dies that are integrated in the package structure. The plurality of trenches are filled with a liquid or a highly thermally conductive metal. When the plurality of trenches are filled with a liquid, a sealing layer is deposited over the plurality of trenches to seal the liquid in the plurality of trenches. The inclusion of the liquid or the highly thermally conductive metal in the substrates increases the heat dissipating ability and improve heat distribution in the package structure.

(9) The various aspects of the present disclosure will now be described in more detail with reference to the figures. In that regard, FIG. 1 is a flowchart illustrating a method **100** of forming a package structure that includes at least a first die and a second die, according to various aspects of the present disclosure. Method **100** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **100**. Additional steps can be provided before, during and after method **100**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all steps are described herein in detail for reasons of simplicity. Method **100** is described below in conjunction with FIG. 2-17, which are fragmentary cross-sectional views of a first workpiece **200** at different stages of fabrication according to various embodiments of method **100**. For avoidance of doubts, the X, Y and Z directions in FIGS. 2-16 are perpendicular to one another. Throughout the present disclosure, unless expressly otherwise described, like reference numerals denote like features.

(10) Referring to FIGS. 1 and 2, method **100** includes a block **102** where a first workpiece **200** that includes a first die **250**, a second die **260** and a gap fill layer **204** is received. In the embodiments represented in FIG. 2, the first workpiece **200** includes a first die **250** and a second die **260** mounted on a first carrier substrate **202**. It should be understood that the first workpiece **200** shown in FIG. 2 is only an example and the first workpiece **200** may include only one die or more than two dies depending on the design needs. Each of the first die **250** and the second die **260** may be a logic die, a system-on-chip (SOC) die, a memory die (e.g., a dynamic random access memory (DRAM) die). The first die **250** includes a first substrate **252**, a first transistor **254** formed on the first substrate **252**, a first through substrate via (TSV) **256** extending through the first substrate **252**, and a first interconnect structure **258**. The first interconnect structure **258** is disposed on a front surface of the first substrate **252** and the first die **250** is flipped upside down to be mounted on the first carrier substrate **202**. It is noted that the first die **250** includes more transistors than just the first transistor **254** and more TSVs than just the first TSV **256**. For ease of illustration and explanation, figures of the present disclosure only show the first transistor **254** and the first TSV **256**. Similarly, the second die **260** includes a second substrate **262**, a second transistor **264** formed on the second substrate **262**, a second through substrate via (TSV) **266** extending through the second substrate **262**, and a second interconnect structure **268**. The second interconnect structure **268** is disposed on a front surface of the second substrate **262** and the second die **260** is flipped

upside down to be mounted on the first carrier substrate **202**. It is noted that the second die **260** includes more transistors than just the second transistor **264** and more TSVs than just the second TSV **266**. For ease of illustration and explanation, figures of the present disclosure only show the second transistor **264** and the second TSV **266**.

(11) In some embodiments, both the first substrate **252** and the second substrate **262** include silicon (Si). Alternatively, the first substrate **252** and the second substrate **262** may include a compound semiconductor, such as silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor, such as silicon germanium (SiGe), GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Alternatively, the first substrate **252** and the second substrate **262** may be semiconductor-on-insulator substrates, such as a silicon-on-insulator (SOI) substrates, silicon germanium-on-insulator (SGOI) substrates, or germanium-on-insulator (GeOI) substrates. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. Both the first substrate **252** and the second substrate **262** can include various doped regions (not shown) depending on design requirements. In some implementations, the first substrate **252** and the second substrate **262** include p-type doped regions (for example, p-type wells) doped with p-type dopants, such as boron (for example, BF.sub.2), indium, other p-type dopant, or combinations thereof as well as n-type doped regions (for example, n-type wells) doped with n-type dopants, such as phosphorus (P), arsenic (As), other n-type dopant, or combinations thereof. The various doped regions can be formed directly on and/or in the first substrate **252** and the second substrate **262**, for example, to provide a p-well structure, an n-well structure, a dual-well structure, a raised structure, or combinations thereof. An ion implantation process, a diffusion process, and/or other suitable doping process can be performed to form the various doped regions.

(12) The first transistor **254** and the second transistor **264** may be planar transistors or multi-gate transistors, such as fin-like field effect transistors (FinFETs) or gate-all-around (GAA) transistors. A planar transistor includes a gate structure that may induce a planar channel region along one surface of its active region, hence its name. A FinFET includes a fin-shaped active region arising from a substrate and a gate structure disposed over a top surface and sidewalls of the fin-shaped active region. A GAA transistor includes at least one channel member extending between two source/drain features and a gate structure that wraps completely around the at least one channel member. Because its gate structure wraps around the channel member, a GAA transistor may also be referred to as a surrounding gate transistor (SGT). Depending on the shapes and orientation, a channel member in a GAA transistor may be referred to as a nanosheet, a semiconductor wire, a nanowire, a nanostructure, a nano-post, a nano-beam, or a nano-bridge. In some instances, a GAA transistor may be referred to by the shape of the channel member. For example, a GAA transistor having one or more nanosheet channel member may also be referred to as a nanosheet transistor or a nanosheet FET.

(13) Referring still to FIG. 2, the first interconnect structure **258** and the second interconnect structure **268** may each include eight (8) to nineteen (19) metal layers. Each of the metal layers may include an etch stop layer (ESL) and an intermetal dielectric (IMD) layer disposed on the ESL. The ESL may include silicon carbide, silicon nitride or silicon oxynitride. The IMD layers may include silicon oxide, tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass (USG), or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silicate glass (FSG), phosphosilicate glass (PSG), boron doped silicate glass (BSG), low-k dielectric material, other suitable dielectric material, or combinations thereof. Example low-k dielectric materials include carbon doped silicon oxide, Xerogel, Aerogel, amorphous fluorinated carbon, benzocyclobutene (BCB), or polyimide. Each of the metal layers includes a plurality of vertically extending vias and horizontally metal lines. The first interconnect structure **258** includes first top metal features **290** of the surface adjacent the first carrier substrate **202**. The second interconnect structure **268** include

second top metal features **292** adjacent the surface adjacent to the first carrier substrate **202**. The first top metal features **290** and the second top metal features **292** may also be referred to as first contact pads **290** and second contact pads **292**, respectively. The contact vias, metal lines in the first interconnect structure **258** and the second interconnect structure **268** may include copper (Cu), tantalum (Ta), nickel (Ni), cobalt (Co), aluminum (Al), or a combination thereof. In one embodiment, the contact vias and metal lines include copper (Cu). The top metal features (i.e., **290** and **292**) may include copper (Cu), aluminum (Al), or an alloy thereof. In one embodiment, the top metal features (i.e., **290** and **292**) may include an alloy of aluminum and copper. While not explicitly shown, the contact vias, metal lines and top metal features may further include a barrier layer to interface the oxygen-containing IMDs. The barrier layer may include titanium nitride (TiN), tantalum nitride (TaN), manganese nitride (MnN), or other transition metal nitride.

(14) In some embodiments represented in FIG. 2, the first TSV **256** extends through the first substrate **252** and the second TSV **266** extends through the second substrate **262**. The first TSV **256** and the second TSV **266** may include copper (Cu), tantalum (Ta), nickel (Ni), cobalt (Co), aluminum (Al), or a combination thereof. In one embodiment, the first TSV **256** and the second TSV **266** include copper (Cu) and may be formed using an electroplating process. In an example process, a seed layer is first deposited using physical vapor deposition (PVD) or chemical vapor deposition (CVD) and then a metal fill layer is deposited using electroplating. While not explicitly shown in the figures, a barrier layer may be formed to space the metal fill layer of the first TSV **256** and the second TSV **266** from the ESLs and IMDs in the first interconnect structure **258** or the second interconnect structure **268**. The barrier layer may include titanium nitride (TiN), tantalum nitride (TaN), manganese nitride (MnN), or other transition metal nitride.

(15) As shown in FIG. 2, the first workpiece **200** may also include a gap fill layer **204** and a photoresist layer **206** over the gap fill layer **204**. The gap fill layer **204** functions to fill gaps between the first die **250** and the second die **260** as well as around the first die **250** and the second die **260**. The photoresist layer **206** may be used to selectively remove the gap fill layer **204** over the first die **250** and the second **260** to prevent formation of uneven surfaces during subsequent planarization processes. In some embodiments, the gap fill layer **204** includes silicon oxide and may be deposited using chemical vapor deposition (CVD), flowable CVD (FCVD), or spin-on coating. The first carrier substrate **202** may include silicon or glass. As will be described further below, the first carrier substrate **202** may be removed at a later stage and will not become part of the final package structure.

(16) Referring to FIGS. 1 and 3-6, method **100** includes a block **104** where the gap fill layer **204** is removed from surfaces of the first die **250** and the second die **260**. Block **104** includes multiple steps to remove the gap fill layer **204** over the first die **250** and the second **260** without forming recesses in the remaining gap fill layer **204**. Block **104** first patterns the photoresist layer **206** using photolithography. In an example process, the photoresist layer **206** may undergo a pre-exposure baking process, exposure to radiation reflected from or transmitted through a photomask, a post-exposure baking process, and developing process, so as to form a patterned photoresist layer **206** as shown in FIG. 3. The patterned photoresist layer **206** includes a first opening **2060** that exposes the gap fill layer **204** directly over the first die **250** and a second opening **2062** that exposes the gap fill layer **204** directly over the second die **260**. Referring then to FIG. 4, the patterned photoresist layer **206** is applied as an etch mask to etch the gap fill layer **204**. As shown in FIG. 4, the etching of the gap fill layer **204** is performed until the first substrate **252** of the first die **250** and the second substrate **262** of the second die **260** are exposed. The etch process of the gap fill layer **204** may be a dry etch process that includes use of an inert gas (e.g., Ar) a fluorine-containing gas (e.g., CF₄, C₂F₆, SF₆ or NF₃), other suitable gases and/or plasmas, and/or combinations thereof. After the etching of the gap fill layer **204**, the patterned photoresist layer **206** may be removed by ashing or selective etching, as shown in FIG. 5. After the removal of the patterned photoresist layer **206**, the first workpiece **200** is planarized using, for example, a chemical

mechanical polishing (CMP) process, to provide a substantial planar top surface, as shown in FIG. 6. It is noted that because the planarization process tends to remove the gap fill layer **204** faster than it does the first substrate **252** or the second substrate **262**, performing a planarization directly to the first workpiece **200** shown in FIG. 2 may result in recesses between and around the first die **250** and the second die **260**. That is why block **104** includes the aforementioned steps to planarize the first workpiece **200**.

(17) Referring to FIGS. 1 and 7, method **100** includes a block **106** where a first substrate **252** and a second substrate **262** are selectively recessed. In some embodiments, the selective recessing may be implemented by a dry etch process that includes use of oxygen-containing gas (e.g., O.sub.2), an inert gas (e.g., Ar) a fluorine-containing gas (e.g., CF.sub.4, C.sub.2F.sub.6, SF.sub.6 or NF.sub.3), other suitable gases and/or plasmas, and/or combinations thereof. As shown in FIG. 7, the selective recessing is performed until top portions of the first TSV **256** and the second TSV **266** rise above top surfaces of the first substrate **252** and the second substrate **262**. As shown in FIG. 7, the gap fill layer **204** may be recessed along with the first substrate **252** and the second substrate **262**.

(18) Referring to FIGS. 1 and 8, method **100** includes a block **108** where a plurality of trenches **272** are formed in the first substrate **252** and the second substrate **262**. Each of the plurality of trenches **272** may include a width W along the X direction and a depth D along the Z direction (i.e., the vertical direction). In some embodiments, the depth D may be between about 1 μm and about 500 μm . The width W of the trenches **272** may have a wide range, depending on the space available for the trenches **272** and how the trenches **272** are formed. The plurality of trenches **272** may be formed using a mechanical dicing saw, a laser dicing saw, or a dry etch process. In the depicted embodiments, the first die **250** and the second **260** are square or rectangular in shape from a top view (i.e., along the Z direction) and may have a width along the Y direction. The trenches **272** in FIG. 8 also extend lengthwise along the Y direction. The length of the trenches **272** extends a portion of the width of the first die **250** and the second die **260**. In some embodiments, one or more of the trenches **272** may extend substantially the entire width of the first die **250** and the second die **260**. That is, one or more of the plurality of trenches **272** may traverse or span substantially the entire Y-direction width of the first die **250** and the second die **260**.

(19) Referring to FIGS. 1 and 9, method **100** includes a block **110** where a liquid **274** is injected into the plurality of trenches **272**. In some embodiments, the liquid **274** is water. In some alternative embodiments, the liquid **274** may be a liquid suitable for high heat flux immersion cooling. In these alternative embodiments, the liquid **274** includes several attributes that make it suitable for use in the semiconductor structure of the present disclosure. For example, the liquid **274** is dielectric and has low dielectric constant to prevent leakage and parasitic capacitance. In some instances, the liquid **274** may have low surface tension so it can better wet the surfaces of the trenches **272**. A suitable liquid **274** should also have high boiling point or low vapor pressure to prevent device damage due to high vapor pressure. Examples of the liquid **274** for these alternative embodiments may include hydrocarbons or fluorocarbons. Example hydrocarbons include polyether. Example fluorocarbons include perfluoropolyether (PFPE) and 1-butyl-3-methylimidazolium hexafluorophosphate (BMIM-PF.sub.6). At block **110**, the liquid **274** may be atomized and injected over the first workpiece **200** to fill the trenches **272**. Alternatively, the liquid **274** may be deposited over the trenches **272** using spin-on coating. In some alternative embodiments, the liquid **274** is replaced with a highly thermally conductive metal, such as aluminum (Al). When a highly thermally conductive metal is used instead of the liquid **274**, the metal may be deposited using atomic layer deposition physical vapor deposition (PVD) or CVD.

(20) Referring to FIGS. 1 and 10, method **100** includes a block **112** where a sealing layer **276** is deposited over the first workpiece **200** to seal the liquid **274** in the trenches **272**. In some embodiments, the sealing layer **276** may include silicon, silicon nitride, or silicon oxide. In one embodiment, the sealing layer **276** includes silicon nitride. The sealing layer **276** may be deposited using ALD, CVD, or flowable CVD (FCVD). When a highly thermally conductive metal is used at

block **110** instead, operations at block **112** may be omitted. After the deposition of the sealing layer **276**, the first TSV **256** and the second TSV **266** may be completely covered by the sealing layer **276**.

(21) Referring to FIGS. **1** and **11**, method **100** includes a block **114** where the sealing layer **276** is planarized to expose a portion of the first TSV **256** and the second TSV **266**. At block **114**, the first workpiece **200** is planarized to remove excess sealing layer **276** until the sealing layer **276** over top surfaces of the first substrate **252** and second substrate **262** is removed and the first TSV **256** and the second TSV **266** partially rise over the sealing layer **276**. As shown in FIG. **11**, upon conclusion of block **114**, the plurality trenches **272** are substantially filled with the liquid **274** and the sealing layer **276**. The sealing layer **276** and the liquid **274** sealed in the trenches **272** by the sealing layer **276** may be collectively referred to as first heat distribution features **280**. In some embodiments represented in FIG. **11**, because the planarization process may remove the first substrate **252**, the second substrate **262**, and the sealing layer **276** faster than it does the first TSV **256** and the second TSV **266**, top portion of the first TSV **256** and the second TSV **266** may rise above the top planar surface of the workpiece **200** in FIG. **11**. When the liquid **274** is replaced with a highly thermally conductive metal in the alternative embodiments, the plurality of trenches **272** are substantially filled with the metal to form the first heat distribution features **280**.

(22) Referring to FIGS. **1** and **12**, method **100** includes a block **116** where an isolation layer **282** is formed over the first workpiece **200**. Operation at block **116** includes deposition of the isolation layer **282** over the first workpiece **200** and planarization of the isolation layer **282** to expose top surfaces of the first TSV **256** and the second TSV **266**. In some embodiments, the isolation layer **282** may include silicon nitride, silicon oxide, silicon oxynitride, or silicon oxycarbonitride. In the depicted embodiment, the isolation layer **282** includes silicon nitride. The isolation layer **282** may be conformally deposited using CVD or a suitable method. After the deposition of the isolation layer **282**, the isolation layer **282** may be disposed on sidewalls and top surfaces of the first TSV **256** and the second TSV **266**. The deposited isolation layer **282** is then planarized using a chemical mechanical polishing (CMP) process. As shown in FIG. **12**, after the planarization, top surfaces of the first TSV **256** and the second TSV **266** are exposed and substantially coplanar with a top surface of the isolation layer **282**.

(23) Referring to FIGS. **1** and **13**, method **100** includes a block **118** where a bonding layer **285** is formed over the first die **250** and the second die **260**. One of the functions of the bonding layer **285** is to provide an aligned communication interface. The first die **250** and the second die **260** may have different contact point patterns from the dies to be bonded over the first die **250** and the second die **260**. The bonding layer **285** redirects patterns of the contact points of the first die **250** and the second die **260** such that they can align with contact points of the dies that are to be stacked over the first die **250** and the second die **260**. Additionally, the bonding layer **285** allows the first die **250** and the second die **260** to be bonded to the overlying dies by direct bonding. Direct bonding requires a high level of surface planarity and a high density of dummy and functional bonding metal features. The bonding layer **285** is configured to be planar and include necessary dummy metal features. Referring to FIG. **13**, the bonding layer **285** includes an etch stop layer (ESL) **284**, an intermetal dielectric (IMD) layer **286**, and a plurality of metal features **288** disposed in and extending through the ESL **284** and the IMD layer **286**. The ESL **284** may include silicon nitride or silicon oxynitride. The IMD layer **286** may include silicon oxide, tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass (USG), or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silicate glass (FSG), phosphosilicate glass (PSG), boron doped silicate glass (BSG), low-k dielectric material, other suitable dielectric material, or combinations thereof. Example low-k dielectric materials include carbon doped silicon oxide, Xerogel, Aerogel, amorphous fluorinated carbon, benzocyclobutene (BCB), or polyimide. The plurality of metal features **288** may include copper (Cu), tantalum (Ta), nickel (Ni), cobalt (Co), aluminum (Al), a combination thereof, or an alloy thereof. In one embodiment, the plurality of metal features **288**

may include copper (Cu). To form the bonding layer **285**, the ESL **284** is first deposited over the first workpiece **200** using CVD. Then the IMD layer **286** is deposited over the ESL **284** using CVD, flowable CVD (FCVD), or spin-on coating. Openings are then formed through the ESL **284** and the IMD layer **286**. A metal fill layer is then deposited in the openings. After excess metal fill layer is removed using CMP, the plurality of metal features **288** are formed.

(24) Referring to FIGS. **1** and **14**, method **100** includes a block **120** where a second workpiece **300** is bonded to the first workpiece **200** to form a third workpiece **400**. In some embodiments, the second workpiece **300** includes a third die **350** and a fourth die **360** mounted on a second carrier substrate **302** similar to the first carrier substrate **202**. It should be understood that the second workpiece **300** shown in FIG. **14** is only an example and the second workpiece **300** may include only one die or more than two dies depending on the design needs. Each of the third die **350** and the fourth die **360** may be a logic die, a system-on-chip (SOC) die, a memory die (e.g., a dynamic random access memory (DRAM) die). The third die **350** includes a third substrate **352**, a third transistor **354** formed on the third substrate **352**, and a third interconnect structure **358**. The third interconnect structure **358** is disposed on a front surface of the third substrate **352**. It is noted that the third die **350** includes more transistors than just the third transistor **354**. For ease of illustration and explanation, figures of the present disclosure only show the third transistor **354**. Similarly, the fourth die **360** includes a fourth substrate **362**, a fourth transistor **364** formed on the fourth substrate **362**, and a fourth interconnect structure **368**. The fourth interconnect structure **368** is disposed on a front surface of the fourth substrate **362**. It is noted that the fourth die **360** includes more transistors than just the fourth transistor **364**. For ease of illustration and explanation, figures of the present disclosure only show the fourth transistor **364**.

(25) At block **120**, the second workpiece **300** is flipped upside down and bonded to the first workpiece **200** to define a third workpiece **400** or a multi-tier semiconductor structure **400**. Although not explicitly shown in the figures, the second workpiece **300** may also include a bonding layer with metal features corresponding to those in the bonding layer **285**. To ensure a strong bonding between the bonding layers, surfaces of the two bonding layers are cleaned to remove organic and metallic contaminants. In an example process, a sulfuric acid hydrogen peroxide mixture (SPM), a mixture of ammonium hydroxide and hydrogen peroxide (SC1), or both may be used to remove organic contaminants on the bonding layers. A mixture of hydrochloric acid and hydrogen peroxide (SC2) may be used to remove metallic contaminants. Besides cleaning, the bonding surfaces of the two bonding layers may be treated by an argon plasma or a nitrogen plasma to activate the surfaces thereof. After the bonding layers are vertically aligned, an anneal is performed to promote the van der Waals force bonding of the IMD layers as well as the surface-activated bonding (SAB) of the metal features in the bonding layers. In some instances, the anneal includes a temperature between about 200° C. and about 300°.

(26) Referring to FIGS. **1** and **15**, method **100** includes a block **122** where the second carrier substrate **302** is removed and operations at blocks **108**, **110** and **112** are performed to the second workpiece **300**. After the second workpiece **300** is bonded to the first workpiece **200** at block **120**, the second carrier substrate **302** is removed to expose surfaces of the third substrate **352** and the fourth substrate **362**. With the third substrate **352** and the fourth substrate **362** exposed, operations similar to those described at blocks **108**, **110**, and **112** may be performed to the second workpiece **300** to form second heat distribution features **380**. Similar to the first heat distribution features **280**, each of the second heat distribution features **380** includes liquid **374** sealed in a trench by a sealing layer **376**. As described above, in some alternative embodiments, the liquid **374** and the sealing layer **376** may be replaced with a highly thermally conductive metal, such as aluminum (Al). As the materials and formation of the liquid **374** and the sealing layer **376** are similar to those for the liquid **274** and the sealing layer **276**, detailed description thereof are omitted for brevity.

(27) Referring to FIGS. **1**, **16** and **17**, method **100** includes a block **124** where further processes are performed. Such further processes may include bonding the third workpiece **400** to a third carrier

substrate **382**, removal of the first carrier substrate **202**, and formation of bump features. In some embodiments represented in FIG. **16**, an isolation layer **304** similar the isolation layer **282** may be formed over the second workpiece **300**. The third carrier substrate **382** may have the same composition as the first carrier substrate **202**. In some implementations, the third carrier substrate **382** may include silicon or glass. To bond the third carrier substrate **382** to the second workpiece **300**, a silicon-oxide-containing layer is formed over the third carrier substrate **382** and the isolation layer **304** may be directly bonded to the silicon-oxide-containing layer. In an alternative embodiment, a die attached film may be used to bond the third carrier substrate **382** to the second workpiece **300**. In order to form bump features over the front side surface (facing down in FIG. **16**) of the first workpiece **200**, the first carrier substrate **202** is removed using a die separation or debonding process. Reference is now made to FIG. **17**. Before bump features are formed to couple to first contact pads **290** in the first interconnect structure **258** and second contact pads **292** in the second interconnect structure **268**, a first passivation layer **402** and a second passivation layer **404** are deposited over the front surface of the first workpiece **200** (now part of the flipped down third workpiece **400**). In some embodiments, the first passivation layer **402** may include silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof and may be deposited using CVD. The second passivation layer **404** may be a polymer layer. In some instances, the second passivation layer **404** may include an epoxy, polyimide, benzocyclobutene (BCB), polybenzoxazole (PBO), or a suitable polymeric material. The second passivation layer **404** may be deposited using CVD, FCVD, or spin-on coating.

(28) The first passivation layer **402** and the second passivation layer **404** may be collectively referred to as a passivation structure. After the formation of the passivation structure, openings directly over the first contact pads **290** and the second contact pads **292** are formed through the passivation structure. Copper posts **406** are then formed in the openings to couple to the first contact pads **290** and the second contact pads **292**. The copper posts **406** may include copper (Cu), nickel (Ni), cobalt (Co), or a copper alloy. While not explicitly shown in the figure, an under-bump-metallization (UBM) may be formed before the formation of the copper post **406**. The UBM layer may include a barrier layer and a seed layer. The barrier layer may include titanium nitride or tantalum nitride. The seed layer may include copper (Cu), silver (Ag), chromium (Cr), tin (Sn), gold (Au), and combinations thereof. After the formation of the copper posts **406**, bump features **408** are formed on the copper posts **406**. The bump features **408** may include a Sn—Cu alloy, a Sn—Ag alloy, or a Sn—Cu—Ag alloy. The bump features **408** may be Controlled Collapse Chip Connection (C4) connections or microbumps.

(29) In one exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes a first interconnect structure, a first substrate disposed on the first interconnect structure, a second interconnect structure disposed on the first substrate, and a second substrate disposed on the second interconnect structure. The first substrate includes a first plurality of trenches extending from a surface adjacent the second interconnect structure into the first substrate. The first plurality of trenches are filled with a liquid.

(30) In some embodiments, the liquid includes water, hydrocarbons, or fluorocarbons. In some implementations, the second substrate includes a second plurality of trenches extending from a surface away from the second interconnect structure into the second substrate. The second plurality of trenches are filled with the liquid. In some embodiments, the first substrate further includes at least one through-substrate-via (TSV) that extends completely through the first substrate. In some instances, the first plurality of trenches are further filled with a sealing layer and a top surface of the sealing layer is coplanar with a top surface of the first substrate. In some embodiments, the semiconductor structure further includes a dielectric structure surrounding and in contact with the first interconnect structure and the first substrate and an isolation film disposed on the first substrate, the sealing layer, and the dielectric structure. In some instances, the semiconductor structure further includes a dielectric layer disposed on the isolation film, and a plurality of contact

features embedded in the dielectric layer. In some embodiments, the sealing layer includes silicon nitride.

(31) In another exemplary aspect, the present disclosure is directed to a semiconductor structure. The semiconductor structure includes a first die, a bonding layer, and a second die. The first die includes a first interconnect structure, a first substrate disposed on the first interconnect structure, and a first plurality of heat distribution lines disposed in the first substrate. The bonding layer is disposed over the first substrate. The second die is disposed on the bonding layer and includes a second interconnect structure, a second substrate disposed on the second interconnect structure, and a second plurality of heat distribution lines disposed in the second substrate.

(32) In some embodiments, each of the first plurality of heat distribution lines and the second plurality of heat distribution lines includes a liquid and a sealing layer over the liquid. In some embodiments, the sealing layer includes silicon oxide, silicon nitride, or silicon. In some implementations, the bonding layer includes a first plurality of metal features. The second interconnect structure includes a second plurality of metal features. Each of the first plurality of metal features is vertically aligned with one of the second plurality of metal features. In some instances, each of the first plurality of heat distribution lines and the second plurality of heat distribution lines includes a metal. In some embodiments, the metal includes aluminum (Al).

(33) In yet another exemplary aspect, the present disclosure is directed to a method. The method includes receiving a first workpiece that includes a first die mounted on a first carrier substrate, the first die including a first substrate and a through-substrate via (TSV), selectively etching the first substrate until a portion of the TSV rises above the first substrate, forming a first plurality of trenches in the first substrate, injecting a liquid into the first plurality of trenches, depositing a first sealing layer over the first workpiece to seal the liquid in the first plurality of trenches, and after the depositing of the first sealing layer, planarizing the first workpiece to expose the TSV.

(34) In some embodiments, the first sealing layer includes silicon oxide, silicon nitride, or silicon. In some implementations, the forming of the first plurality of trenches includes use of a mechanical saw. In some embodiments, the forming of the first plurality of trenches includes use of a laser saw. In some instances, the method further includes bonding on the first workpiece a second workpiece that includes a second die mounted on a second carrier substrate, the second die including a second substrate, removing the second carrier substrate, after the removing of the second carrier substrate, forming a second plurality of trenches in the second substrate, injecting the liquid into the second plurality of trenches, depositing a second sealing layer over the second workpiece to seal the liquid in the second plurality of trenches, and after the depositing of the second sealing layer, planarizing the second workpiece. In some embodiments, the method further includes after the planarizing the second workpiece, depositing an isolation layer over the second workpiece, and bonding a third carrier substrate to the isolation layer.

(35) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a first interconnect structure; a first substrate disposed on the first interconnect structure; a second interconnect structure disposed on the first substrate; and a second substrate disposed on the second interconnect structure, wherein the first substrate

comprises a first plurality of trenches extending from a surface adjacent the second interconnect structure into the first substrate, wherein the first plurality of trenches is partially filled with a liquid.

2. The semiconductor structure of claim 1, wherein the liquid comprises water, hydrocarbons, or fluorocarbons.

3. The semiconductor structure of claim 1, wherein the second substrate comprises a second plurality of trenches extending into the second substrate from a surface that faces away from the second interconnect structure, wherein the second plurality of trenches is partially filled with the liquid.

4. The semiconductor structure of claim 3, wherein the first substrate further comprises at least one through-substrate-via (TSV) that extends completely through the first substrate.

5. The semiconductor structure of claim 3, wherein the first plurality of trenches is further filled with a sealing layer such that the liquid and the sealing layer substantially fill the first plurality of trenches, wherein a top surface of the sealing layer is coplanar with a top surface of the first substrate.

6. The semiconductor structure of claim 5, further comprising: a dielectric structure surrounding and in contact with the first interconnect structure and the first substrate; and an isolation film disposed on the first substrate, the sealing layer, and the dielectric structure.

7. The semiconductor structure of claim 6, further comprising: a dielectric layer disposed on the isolation film; and a plurality of contact features embedded in the dielectric layer.

8. The semiconductor structure of claim 6, wherein the sealing layer comprises silicon nitride.

9. A semiconductor structure, comprising: a first die comprising: a first interconnect structure, a first substrate disposed on the first interconnect structure, and a first plurality of heat distribution lines disposed in the first substrate; a bonding layer disposed over the first substrate; and a second die disposed on the bonding layer and comprising: a second interconnect structure, a second substrate disposed on the second interconnect structure, and a second plurality of heat distribution lines disposed in the second substrate.

10. The semiconductor structure of claim 9, wherein each of the first plurality of heat distribution lines and the second plurality of heat distribution lines comprises a liquid and a sealing layer over the liquid.

11. The semiconductor structure of claim 10, wherein the sealing layer comprises silicon oxide, silicon nitride, or silicon.

12. The semiconductor structure of claim 9, wherein the bonding layer comprises a first plurality of metal features, wherein the second interconnect structure comprises a second plurality of metal features, wherein each of the first plurality of metal features is vertically aligned with one of the second plurality of metal features.

13. The semiconductor structure of claim 9, wherein each of the first plurality of heat distribution lines and the second plurality of heat distribution lines comprises a metal.

14. The semiconductor structure of claim 13, wherein the metal comprises aluminum (Al).

15. A method, comprising: receiving a first workpiece that includes a first die mounted on a first carrier substrate, the first die comprising a first substrate and a through-substrate via (TSV); selectively etching the first substrate until a portion of the TSV rises above the first substrate; forming a first plurality of trenches in the first substrate; injecting a liquid into the first plurality of trenches; depositing a first sealing layer over the first workpiece to seal the liquid in the first plurality of trenches; and after the depositing of the first sealing layer, planarizing the first workpiece to expose the TSV.

16. The method of claim 15, wherein the first sealing layer comprises silicon oxide, silicon nitride, or silicon.

17. The method of claim 15, wherein the forming of the first plurality of trenches comprises use of a mechanical saw.

18. The method of claim 15, wherein the forming of the first plurality of trenches comprises use of a laser saw.

19. The method of claim 15, further comprising: bonding on the first workpiece a second workpiece that includes a second die mounted on a second carrier substrate, the second die comprising a second substrate; removing the second carrier substrate; after the removing of the second carrier substrate, forming a second plurality of trenches in the second substrate; injecting the liquid into the second plurality of trenches; depositing a second sealing layer over the second workpiece to seal the liquid in the second plurality of trenches; and after the depositing of the second sealing layer, planarizing the second workpiece.

20. The method of claim 19, further comprising: after the planarizing the second workpiece, depositing an isolation layer over the second workpiece; and bonding a third carrier substrate to the isolation layer.
